

Úloha 1.1

Chceme dokázat:

$$\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2)$$

$$\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) = \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2)$$

Využijeme vlastnosti skalárního součinu a skutečnosti, že $2\operatorname{Re}(x) = x + \bar{x}$ a $-2i\operatorname{Im}(x) = x - \bar{x}$:

$$\begin{aligned} \frac{1}{4}(\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2) &= \frac{1}{4}(\langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \\ &\quad - \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{v}, \mathbf{v} \rangle) = \\ &= \frac{1}{4}(2\langle \mathbf{u}, \mathbf{v} \rangle + 2\overline{\langle \mathbf{u}, \mathbf{v} \rangle}) = \operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

$$\begin{aligned} \frac{1}{4}(\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2) &= \frac{1}{4}(\langle \mathbf{u}, \mathbf{u} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \\ &\quad + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle - i\langle \mathbf{u}, \mathbf{v} \rangle + i\overline{\langle \mathbf{u}, \mathbf{v} \rangle} - \langle \mathbf{v}, \mathbf{v} \rangle) = \\ &= \frac{1}{4}(-2i\langle \mathbf{u}, \mathbf{v} \rangle + 2i\overline{\langle \mathbf{u}, \mathbf{v} \rangle}) = \operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle) \end{aligned}$$

PS: Nechť $x = a + ib$, potom $\bar{x} = a - ib$. $\operatorname{Im}(x) = -\frac{1}{2}i(a + ib - a + ib) = b$.